

Computational Dynamics Q – Assignment Q2

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Q2 – INVERSE DYNAMICS

A space shuttle is floating in space and moving its relatively heavy cargo using a robotic arm that is attached to the space shuttle – see Figure 1.

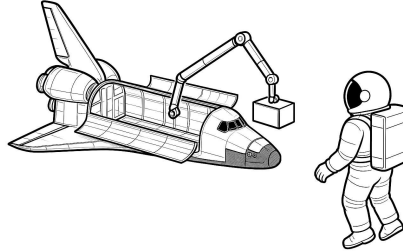


Figure 1

If we want to prevent the space shuttle from rotating or moving relative to an inertial observer – e.g. a floating astronaut – due to the shift in mass while moving the payload, the space shuttle can apply a vectored thrust to counteract it. We want to determine exactly what thrust to apply to achieve this based on an arbitrary motion of the robotic arm.

This will be done using inverse dynamics, i.e. we prescribe the motion of the robotic arm and the space shuttle and then determine the necessary forces to achieve that.

Body 5 / space shuttle

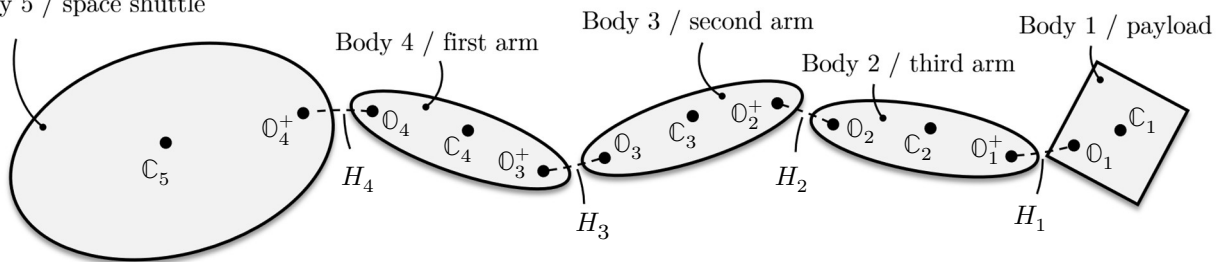


Figure 2

Figure 2 shows a body diagram of the system with the payload being the tip body and the robotic arm consisting of three identical smaller bodies. Because the space shuttle is forced to be still compared to an inertial observer, it itself can be considered an inertial reference frame which qualifies it to be considered as the base body for the scatter-gather algorithms, hence the number of movable bodies is $n = 4$. Hinge 1 and 4 are spherical joints while hinges 2 and 3 are revolute joints. The properties of the bodies are specified in Table 1.

Table 1: Properties of the bodies.

Body, k	Mass, m_k [kg]	CG pos, ${}^k l(\mathbb{O}_k, \mathbb{C}_k)$ [m]	Hinge pos, ${}^k l(\mathbb{O}_k, \mathbb{O}_{k-1}^+)$ [m]	2nd order moment of inertia, $\mathbb{C}_k \mathcal{J}_k$ [kgm ²]
5	N/A	$[0 \ 0 \ 0]^\top$	$[12 \ 4 \ 2]^\top$	N/A
2, 3, 4	400	$[0 \ 0 \ 2.5]^\top$	$[0 \ 0 \ 5]^\top$	$\text{diag}(834, 834, 0.72)$
1	14,520	$[0 \ 0 \ 0.7]^\top$	N/A	$\text{diag}(1.01, 1.51, 0.331) \cdot 10^4$

The size and meaning of the generalized orientation, $\theta(k) \in \mathbb{R}^{r_p(k)}$, and velocity, $\beta(k) \in \mathbb{R}^{r_v(k)}$, will depend on the type of hinge and how we choose to represent the degrees of freedom within them. For this assignment they will be chosen as follows.

- For the spherical hinge, i.e. $\forall k \in \{1, 4\}$, the generalized orientation will be represented by a full unit quaternion such that $\theta(k) \triangleq {}^{k+1}\underline{q}_k \in \mathbb{R}^4$ and the generalized velocity will be the full angular velocity vector such that $\beta(k) \triangleq \omega(k) \in \mathbb{R}^3$.
- For the revolute hinge, i.e. $\forall k \in \{2, 3\}$, the generalized orientation will be the angle between the z -axes of the k 'th and the $(k-1)$ 'th body frames with the x -axes being colinear such that $\theta(k) \in \mathbb{R}$ – see Figure 3. The generalized velocity will in this case be $\beta(k) \triangleq \dot{\theta}(k) \in \mathbb{R}$.

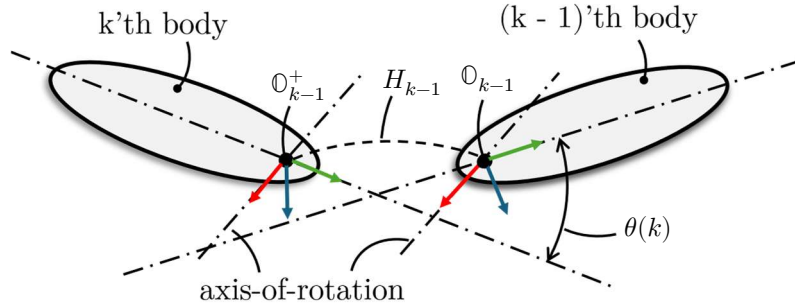


Figure 3: The definition of the angle $\theta(k) \forall k \in \{2, 3\}$ (the revolute hinges). The frames are coincident but split for readability. Red, blue, and green are the x , y , and z axes respectively for their respective frame.

The generalized accelerations, $\gamma(k) \in \mathbb{R}^{r_v(k)}$, will be defined as the time-derivative of the generalized velocities, $\gamma(k) \triangleq \dot{\beta}(k)$. Since all degrees of freedom are non-prismatic, we will not deal with any generalized positions.

Q2.1 Equation of motion

Using the following definitions for the generalized accelerations,

$$\gamma(1) = [-2A \ 0 \ -2A]^\top \cdot 10^{-6} \text{ rad/s}^2 ,$$

$$\gamma(2) = -8A \cdot 10^{-6} \text{ rad/s}^2 ,$$

$$\gamma(3) = 4A \cdot 10^{-6} \text{ rad/s}^2 ,$$

$$\gamma(4) = [5A \quad 0 \quad 6A]^\top \cdot 10^{-6} \text{ rad/s}^2 ,$$

with A being a time, t , dependent amplitude defined as $A \triangleq \sin(2t\pi \cdot 10^{-3})$, define the equation of motion with the state, $S \triangleq [\theta^\top \quad \beta^\top]^\top \in \mathbb{R}^{18}$, consisting of the system-level generalized orientation, $\theta \triangleq \text{col}\{\theta(k)\}_{k=1}^n \in \mathbb{R}^{10}$, And the system-level generalized velocities, $\beta \triangleq \text{col}\{\beta(k)\}_{k=1}^n \in \mathbb{R}^8$, such that it maps $S \rightarrow \dot{S}$.

Hints:

- As in Assignment Q1, the equation of motion should take in a state vector and return the derivative of it but this time the accelerations are given since it is an inverse dynamics problem and there are no generalized positions.
- Define a function that maps $\theta(k) \rightarrow \dot{\theta}(k)$ for any hinge type.

Q2.2 Scatter loop

Define a function that uses the scatter loop algorithm to map the system-level generalized vectors, θ , β , and γ with their corresponding system-level spatial vectors $\underline{q} \triangleq \text{col}\{\underline{q}_k\}_{k=1}^n \in \mathbb{R}^{16}$, $\mathcal{V} \triangleq \text{col}\{\mathcal{V}(k)\}_{k=1}^n \in \mathbb{R}^{24}$, and $\alpha \triangleq \text{col}\{\alpha(k)\}_{k=1}^n \in \mathbb{R}^{24}$.

Hints:

- Mapping between the generalized velocity and the spatial velocity across the hinge is done through a linear hinge map by $\Delta_{\mathcal{V}}(k) = H^\top(k)\beta(k)$ but mapping between generalized orientation and a quaternion can be non-linear and must therefore in the general case be done by a function. Define a separate function that maps $\theta(k) \rightarrow {}^{k+1}\underline{q}_k$ depending on the hinge type. Note that for the revolute hinge, the definition of the generalized orientation fits well with the angle/axis parameter representation. Look at how this representation relates to unit quaternions.

Q2.3 Gather loop

Define a function that implements the gather loop algorithm to map the spatial vectors $\underline{q}$, $\mathcal{V}$, and α with the system-level spatial force $\mathbf{f} \triangleq \text{col}\{\mathbf{f}(k)\}_{k=1}^n \in \mathbb{R}^{24}$.

Q2.4 Simulation

Using the previously defined functions and the initial conditions for the generalized orientation and velocity as,

$$\theta|_{t=0} = [0 \ 0 \ 0 \ 1 \ \pi/4 \ \pi/4 \ 0 \ 0 \ 0 \ 1]^\top ,$$

$$\beta|_{t=0} = [0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0]^\top ,$$

simulate the system for 1,000 sec to plot the spatial forces that the surrounding space must act on the space shuttle for it to not move relative to an inertial observer. Represent this force translated and rotated to the CG of the space shuttle.

Hints:

- Integrate the equation of motions to determine the generalized accelerations and then afterwards loop through each timestep with the scatter and gather loop to determine the spatial forces.
- The final spatial force for $k = 4$ is the spatial force of the arm acting on the space shuttle. Since the space shuttle isn't connected any further, this must therefore be the force that the space shuttle acts on the surrounding space. When using the same formula as in the gather algorithm to translate the force to body 5, you will get the force of the surrounding space acting on the space shuttle.
- The expected movement of the plots can be seen in Figure 4. If you want to replicate this plot (optional) consider creating a second scatter function that defines the rotation matrices, hinge- and centroidal positions as you propagate outwards.
- The final plot should resemble that of Figure 5.

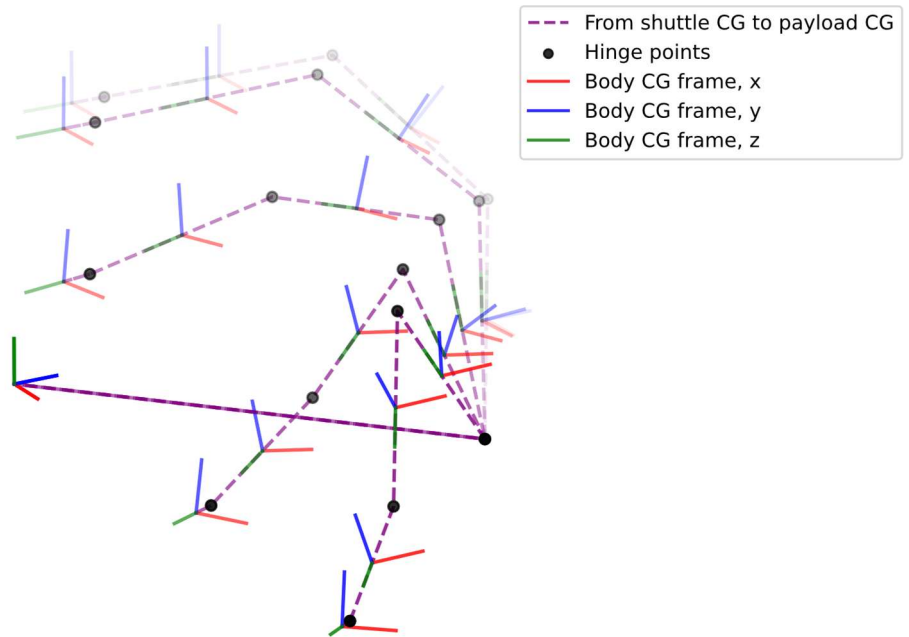


Figure 4: The position of the bodies through time. The most transparent is the initial condition while the least is the final.

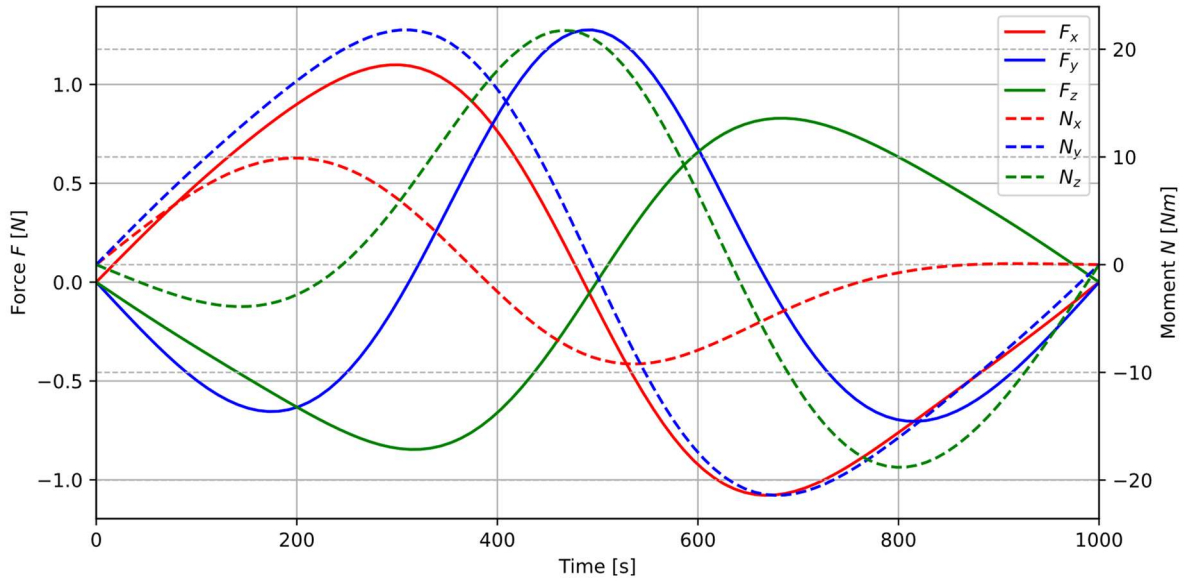


Figure 5: The spatial forces of the surrounding space acting on the space shuttle translated and rotated to the CG of the space shuttle.

Q2.5 Interpretation of the generalized accelerations (extra / optional)

Taking the derivative of a vector with respect to time in a rotating frame doesn't give you an interpretable value. For an example, if $l(k)$ is the position vector and $\nu(k)$ is the linear velocity vector, with respect to an inertial reference frame, of the k 'th body frame, then,

$$\frac{d_k l(k)}{dt} \neq \nu(k) ,$$

as long as the k 'th body frame isn't an inertial frame. This isn't intuitive. For this assignment we only use angular velocities as the generalized velocities which are represented in a rotating body frame while the generalized accelerations are the time derivative of that. If $\alpha(k)$ is the angular acceleration of the k 'th body frame, show that,

$$\frac{d_k \omega(k)}{dt} = \alpha(k) ,$$

is indeed true for this specific vector and that the generalized accelerations for this assignment therefore does reflect an interpretable value. Would this still be the case if the hinges included prismatic joints?

Hints:

- For any vector x the notation $\frac{d_k x(k)}{dt}$ means 'the derivative of x as seen from k with respect to time', i.e. $\frac{d_k x(k)}{dt} \triangleq \frac{d({}^k x(k))}{dt}$.